

Numbering systems

Decimal numbers

We use the decimal system to represent numbers:

- The system uses 10 digits (hence “decimal”), from 0 to 9
- 10 is called the *base*
- Numbers are represented by using *powers of the base*
- For example, the number **1235** is:

$$\begin{array}{ccccccc} & 1000 + & & 200 + & & 30 + & & 5 = \\ \mathbf{1} \times 1000 + & & \mathbf{2} \times 100 + & & \mathbf{3} \times 10 + & & \mathbf{5} \times 1 = \\ \mathbf{1} \times 10^3 + & & \mathbf{2} \times 10^2 + & & \mathbf{3} \times 10^1 + & & \mathbf{5} \times 10^0 \end{array}$$

Binary numbers

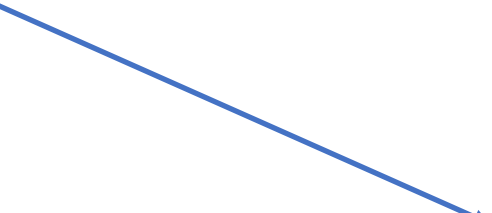
Computers store data in bits, which can take only two values: on/off, true/false and, for numbers, 0/1

- Instead of 10 digits to represent numbers, computers have two: 0 and 1
- We can represent numbers using *base 2*
 - Like decimal, with powers of 2
- Next, an example: convert the binary number **1010011** to decimal.

Binary numbers

The binary number **1010011** is, in decimal:

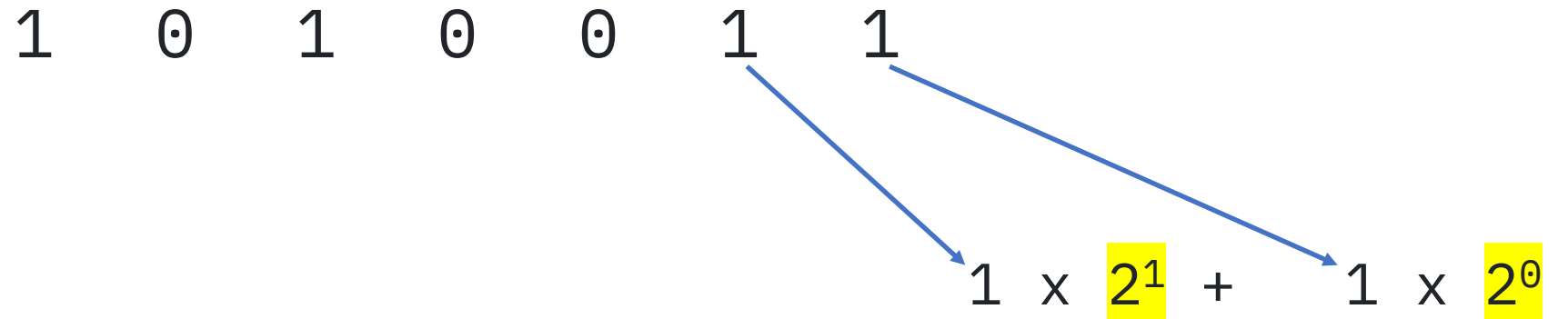
1 0 1 0 0 1 1



1 x 2⁰

Binary numbers

The binary number **1010011** is, in decimal:



Binary numbers

The binary number **1010011** is, in decimal:

$$\begin{array}{ccccccc} 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ & & & & \searrow & \searrow & \searrow \\ & & & & 0 & 1 & 1 \\ & & & & \times & \times & \times \\ & & & & 2^2 & 2^1 & 2^0 \end{array}$$

$0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

Binary numbers

The binary number **1010011** is, in decimal:

1 0 1 0 0 1 1

0 × 2³ + 0 × 2² + 1 × 2¹ + 1 × 2⁰

Binary numbers

The binary number **1010011** is, in decimal:

The diagram illustrates the conversion of the binary number 1010011 to its decimal value. The binary digits are arranged in a row: 1, 0, 1, 0, 0, 1, 1. Below each digit, a blue arrow points to its corresponding term in a sum. The terms are: 1 x 2⁴, 0 x 2³, 0 x 2², 1 x 2¹, and 1 x 2⁰. The powers of 2 are highlighted in yellow boxes. The sum is written as: 1 x 2⁴ + 0 x 2³ + 0 x 2² + 1 x 2¹ + 1 x 2⁰.

$$1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

Binary numbers

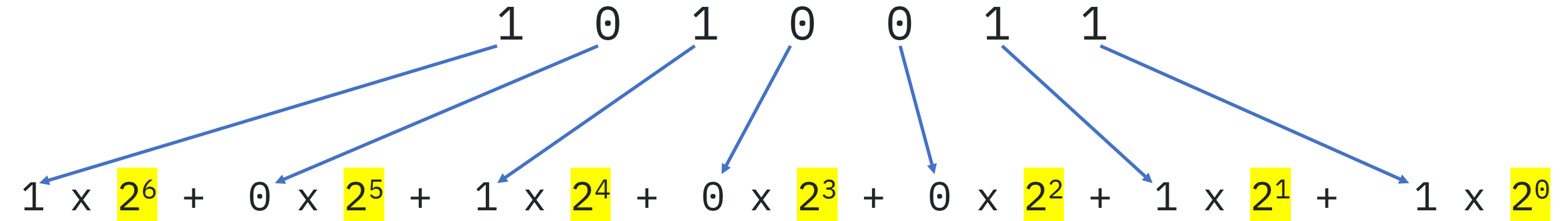
The binary number **1010011** is, in decimal:

The diagram illustrates the conversion of the binary number **1010011** to its decimal expansion. The binary digits are aligned with their corresponding powers of 2, with arrows indicating the mapping:

$$0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

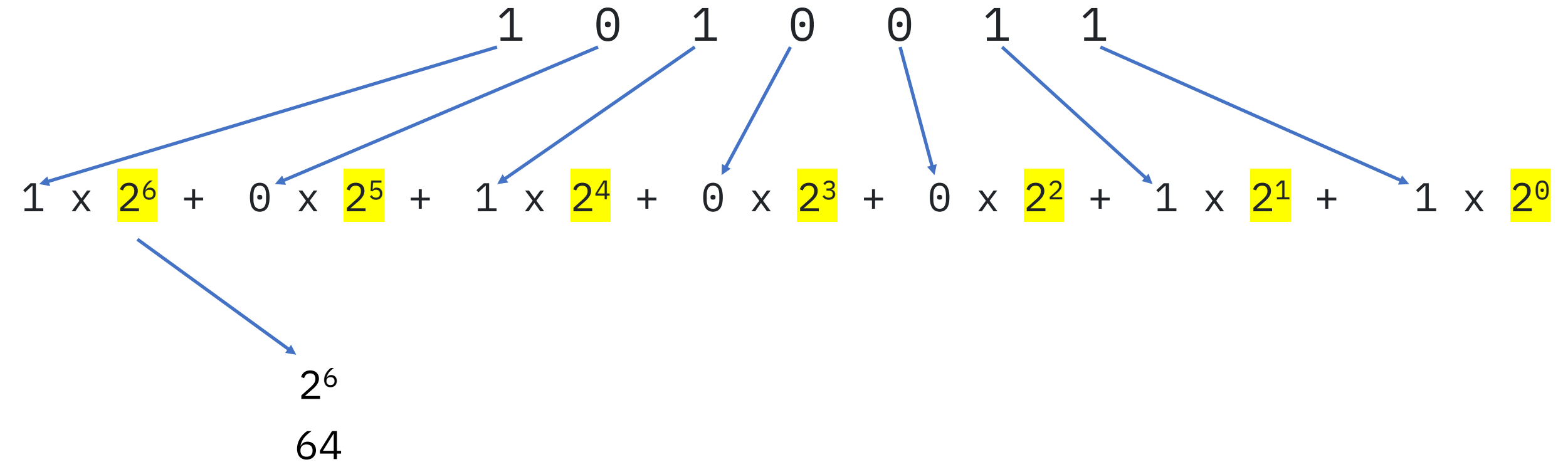
Binary numbers

The binary number **1010011** is, in decimal:



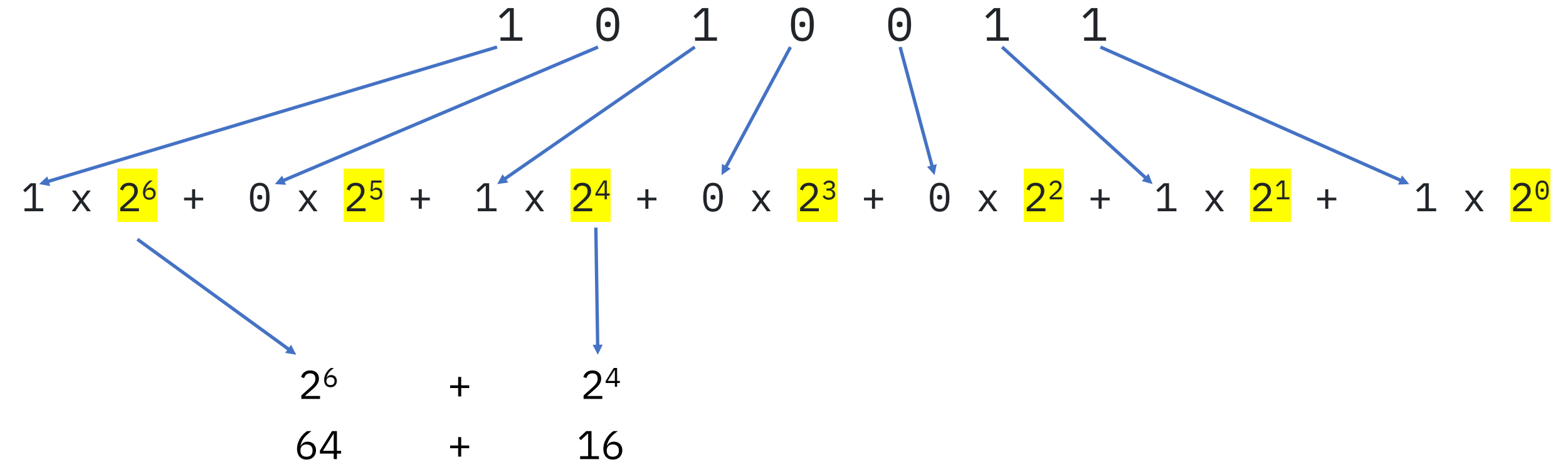
Binary numbers

The binary number **1010011** is, in decimal:



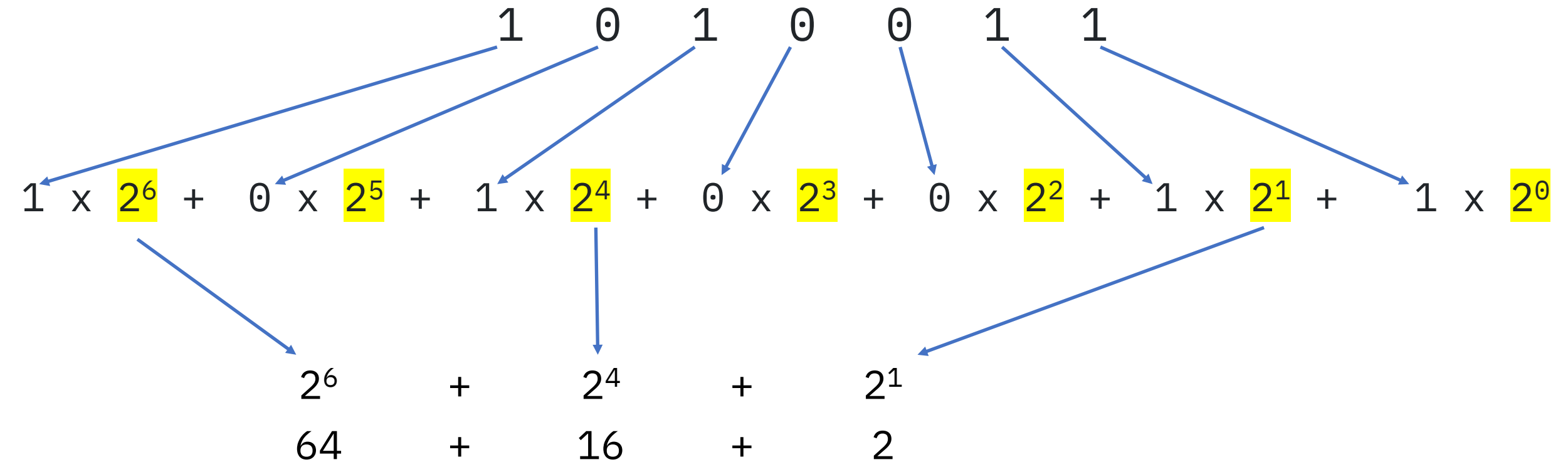
Binary numbers

The binary number **1010011** is, in decimal:



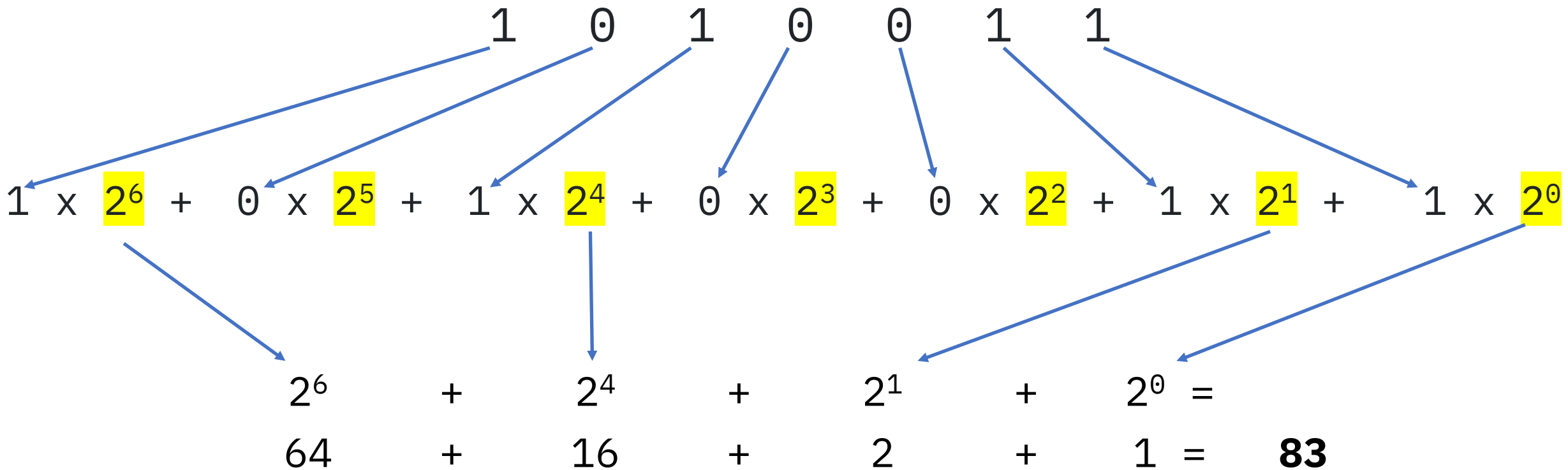
Binary numbers

The binary number **1010011** is, in decimal:



Binary numbers

The binary number **1010011** is, in decimal:

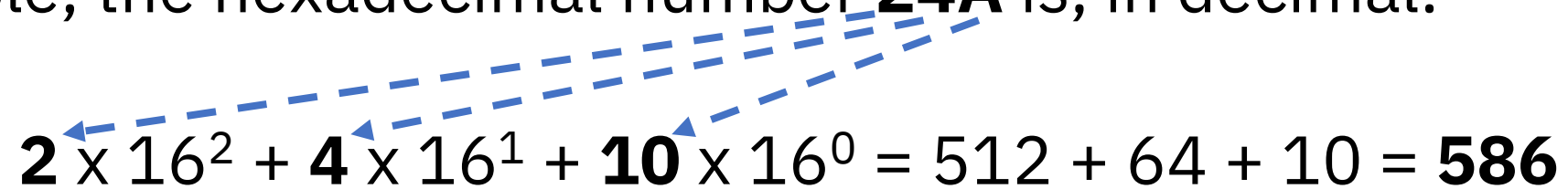


The problem with binary

- Binary numbers have a problem: they can be too long
- Take for example, the number **1235**. In binary, it is:
10011010011 (You can verify this by calculating $2^{10} + 2^7 + 2^6 + \dots$)
- It takes 11 binary digits to represent 4 decimals!
- The solution? The hexadecimal (base 16) numbering system, presented next

Hexadecimal numbers

- As we've seen, we can use any base to represent numbers
- Base 16 (hexadecimal, abbreviated "hex") is very useful
 - We'll see why soon
- To represent numbers in base 16 we need... 16 digits. We use:
 - 0 to 9, same as decimal
 - A to F, to represent 10 to 15
- For example, the hexadecimal number **24A** is, in decimal:


$$\mathbf{2} \times 16^2 + \mathbf{4} \times 16^1 + \mathbf{10} \times 16^0 = 512 + 64 + 10 = \mathbf{586}$$

Hexadecimal numbers

- Hexadecimal numbers are useful for representing binary data
- We can represent groups of 4 bits with a single hexadecimal digit:

Decimal	0	1	2	3	4	5	6	7
Binary	0000	0001	0010	0011	0100	0101	0110	0111
Hexadecimal	0	1	2	3	4	5	6	7

Decimal	8	9	10	11	12	13	14	15
Binary	1000	1001	1010	1011	1100	1101	1110	1111
Hexadecimal	8	9	A	B	C	D	E	F

Next, an example: represent decimal **1235** in hexadecimal

Hexadecimal numbers - example

- Recall that decimal **1235** is, in binary, **10011010011**
- To represent it in hexadecimal:
 - We first arrange it in groups of 4 bits from the right: **100 1101 0011**
 - then convert each 4-bit group to its corresponding hex digit:

Binary	0100	1101	0011
Hexadecimal	4	D	3

- Binary **10011010011** is **4D3** in hexadecimal
 - Much shorter!
- Next: how to convert a decimal number to hexadecimal, step by step